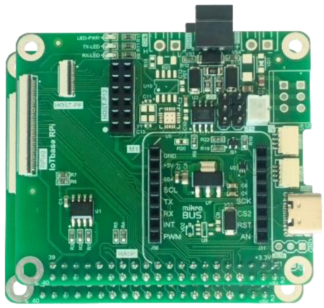
IoTbase RPi

Base module for Raspberry Pi

The **IoTbase RPi** module (version 3.04) is designed to connect **Raspberry Pi** with mezzanines from the **IoTextra** series, **mikroBUS™** modules, and to implement external interfaces for **Raspberry Pi**

A 12-pin **HOST** connector is used to connect mezzanines; it carries the **I²C** bus signals, 8 **GPIO** signals, and 5VDC power with ground.

A **mikroBUS™** slot is available. It can accept **Click©** modules and other modules using this bus, such as System-on-Module (SoM) **SAMD21 Lite**, **BM83x** (BLE based on Nordic chips), or others.



FFC connectors are also provided for combining the **Raspberry Pi** bus across multiple modules (inter-module connections) and for repeating the **HOST** connector signals.

The **I²C** interface is the one most commonly used when building devices and instruments based on this module. For this purpose, the module has two **Qwiic©** connectors for connecting to other modules over the **I²C** bus, through which external devices and sensors can be attached. Pull-up resistors for **I²C** can be connected if needed.

The module implements a non-isolated **RS485** interface.

On the **IoTbase RPi** module, an **EEPROM** is installed for storing configuration and user data. EEPROM capacity - 8 Kbit or 64 Kbit.

The module's input power is non-isolated, in the range from 9VDC to 36VDC. This power can be supplied through a 2-pin terminal block with a 3.5mm pitch or through the **USB** connector.

Module size - **65 x 56 mm**. The module's form factor matches the popular **Raspberry Pi A** board format, which significantly simplifies its use with any modules equipped with a 40-pin connector.

Main application areas of the module:

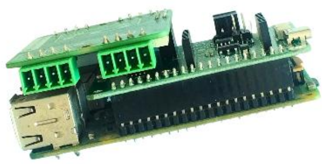
- Industry and transport
- Field devices and PLC
- Ventilation, heating and lighting
- Consumer electronics
- Environmental monitoring
- Smart home
- Power on/off switching
- Gaming applications

QUICK START

The **IoTbase RPi** module with an **IoTextra** series mezzanine installed can easily be used as a HAT for **Raspberry Pi**, thanks to the 40-pin connector it has.

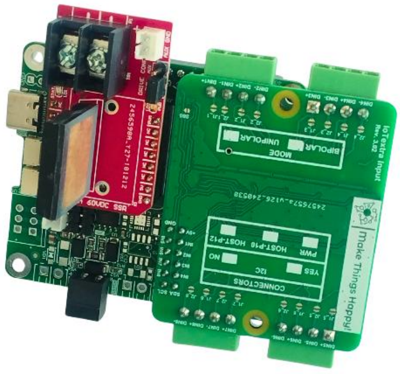
If the module is powered through the **USB** or **PWR** connector, then nothing needs to be done with the jumpers on the **IoTbase RPi** module in this case (use their default states). If the module is powered entirely from **Raspberry Pi**, then jumper **SB20** must be closed.

The following photos show the **IoTbase RPi** module with **Raspberry Pi**, as well as with various mezzanines and **mikroBUS™** modules installed:



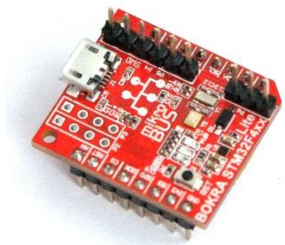
IoTbase RPi with **IoTextra**, installed into the 40-pin **Raspberry Pi** connector

IoTbase RPi with **IoTextra Octal2**



IoTbase RPi with **IoTextra Input and 60VDC SSR**

You can also install one of our own **SoM** modules into the **mikroBUS™** slot, or a wireless communication board, as well as one of our modules with a **mikroBUS™** interface:



IoT STM32F4xx Lite

IoT LPC824 Lite

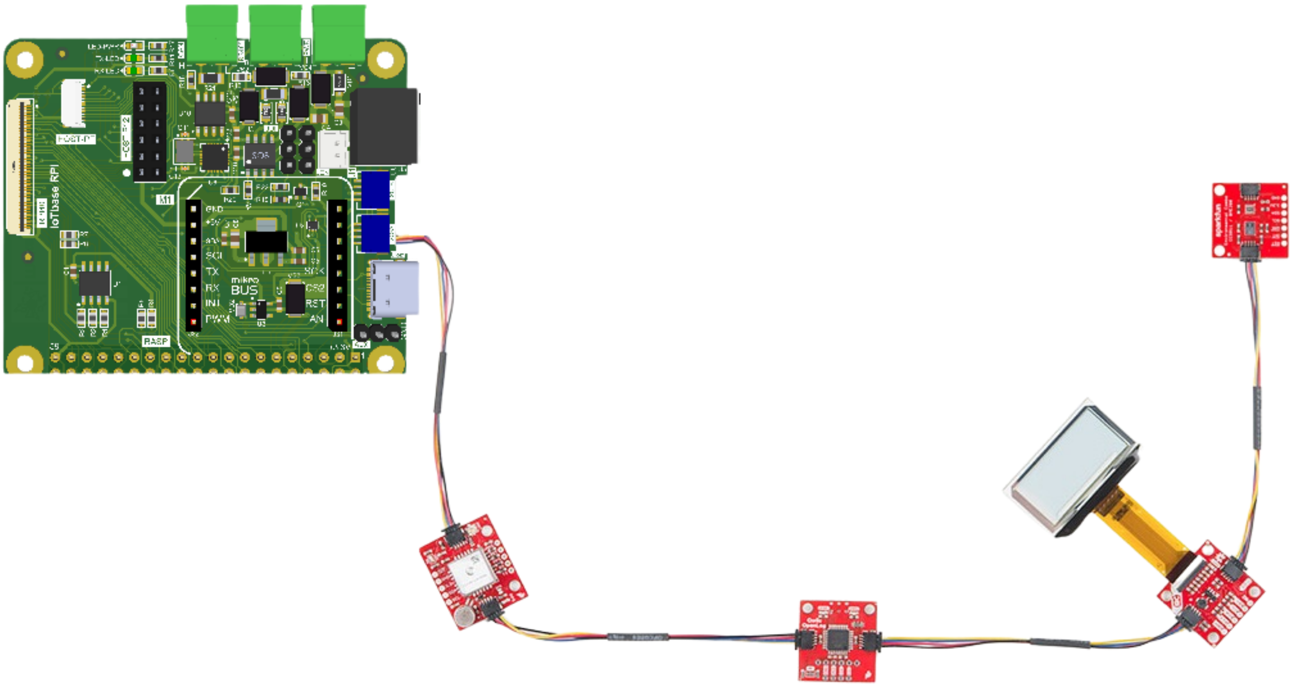
IoT BM833

IoT mikroBUS™ 60VDC SSR

The company [MikroElektronika] produces more than 1000 modules with the **mikroBUS™** interface — **Click®** modules:

To install **Click®** modules into the **mikroBUS™** slot, nothing needs to be done with the jumpers on the **IoTbase RPi** module (use their default states).

Numerous sensors, peripherals, and modules compatible with Qwiic© can also be easily connected to **IoTbase RPi** through the I^{2}C connectors:

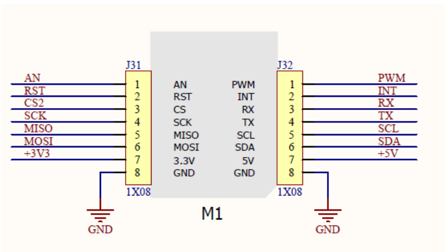


DESCRIPTION

The correspondence between the **mikroBUS™** bus signals of the **IoTbase RPi** module and **Raspberry Pi** is shown in the table below:

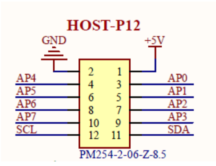
		Pin No.			
DC Power	3.3V	1	2	5V	
SDA	GPIO2	3	4	5V	
SCL	GPIO3	5	6	GND	
AN1 / RSE	GPIO4	7	8	GPIO14	TX
	GND	9	10	GPIO15	RX
RST	GPIO17	11	12	GPIO18	AN
INT	GPIO27	13	14	GND	
INT1	GPIO22	15	16	GPIO23	RST1
DC Power	3.3V	17	18	GPIO24	AP4
MOSI	GPIO10	19	20	GND	
MISO	GPIO9	21	22	GPIO25	CS2
SCK	GPIO11	23	24	GPIO8	CS
	GND	25	26	GPIO7	CS1
SDA1	GPIO0	27	28	GPIO1	SCL1
AP0	GPIO5	29	30	GND	
AP1	GPIO6	31	32	GPIO12	PWM
PWM1	GPIO13	33	34	GND	
AP2	GPIO19	35	36	GPIO16	AP5
AP3	GPIO26	37	38	GPIO20	AP6
	GND	39	40	GPIO21	AP7

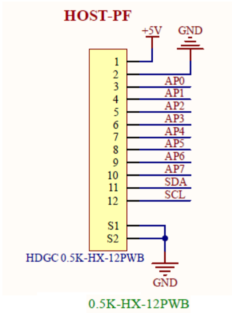
The **IoTbase RPi** module has a **mikroBUS™** slot:



The **HOST-P12** connector for communication with application mezzanines is mounted on the top-side of the **IoTbase RPi** module. The **AP0-AP7** GPIO signals are generated by **Raspberry Pi**. A copy of these signals is available on the **HOST-PF** connector.

Connector structure:

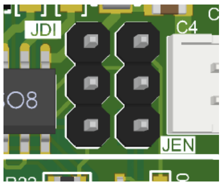




RS485

The module implements a non-isolated **RS485** interface based on the **SP3485EN-L/TR** transceiver (half-duplex). The TX and RX lines are connected to the transceiver through 20Ω resistors; transmit and receive activity is indicated by two green LEDs — **TX-LED** and **RX-LED**.

Operating mode: automatic or software-controlled



RS485 is a half-duplex bus: at any given moment the module is either transmitting or receiving, and something must decide when to switch between these states. **IoTbase RPi** supports two ways of doing this switching. The choice is made by the position of two jumpers on the top-side of the board — **JEN** and **JDI** (next to U8/C4).

- **Automatic mode** — the direction switches in hardware, without any involvement from **Raspberry Pi**: a small transistor circuit monitors the TX line itself and enables the transmitter whenever there is something to send. This is the simplest option — it works out of the box, nothing needs to be configured in software.
  Set **JEN** to position **2-3** and **JDI** to position **2-3**.
- **Software-controlled mode** — the direction is controlled by **Raspberry Pi** itself via the **GPIO4** signal. This mode is needed when the application needs to explicitly decide for itself when the bus is busy transmitting and when it is free to receive.
  Set **JEN** to position **1-2** and **JDI** to position **1-2**.
  In addition, jumper **SB5** (bottom-side) must be closed — it connects **GPIO4** to this control circuit.

Both jumpers must always be moved together, to the same side — mixing positions (e.g., **JEN** on 1-2 while **JDI** is on 2-3) is not allowed; the mode will not work.

Line matching and protection

Bias resistors (fail-safe biasing) pull the A and B lines to defined levels so that the bus does not "float" in the idle state and generate false signals. Diodes protect the lines from voltage transients.

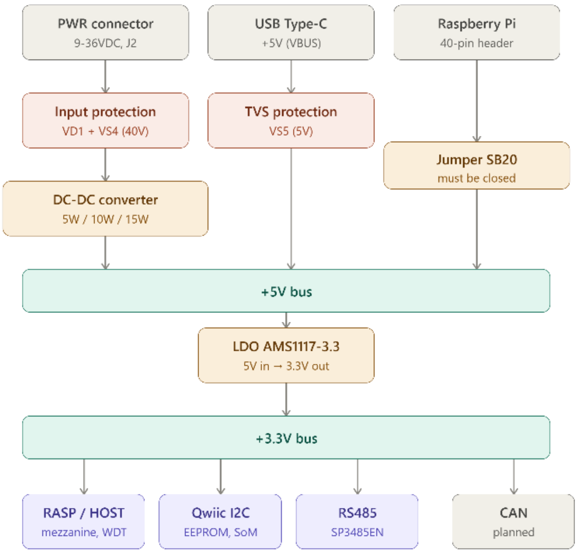
If this module is installed at the physical end of the RS485 bus, the **120Ω** termination resistor needs to be enabled — this is done by closing jumper **SB6**. If the module is located in the middle of the bus (between other devices), **SB6** must remain open.

Connection

The **RS485** output connector is a 2-pin terminal block (A and B). There is no separate GND pin on the connector — the common wire is carried through the power supply.

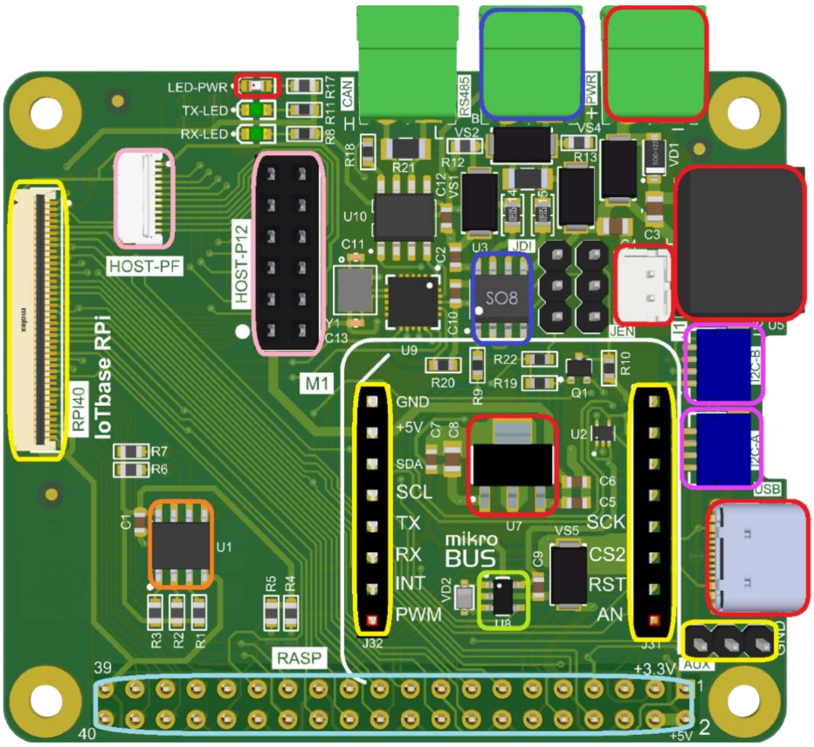
POWER SUPPLY

The power supply organization in the module is shown in the following picture:



LAYOUT

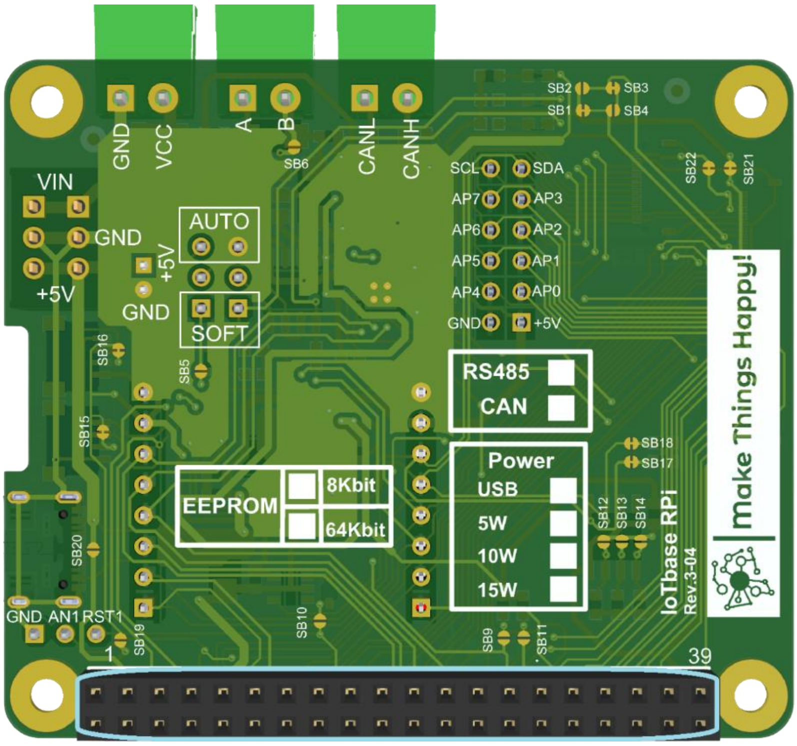
Below is the layout of the elements on the top-side of the **IoTbase RPi** module:



In this picture:

- **Red** highlights power-related elements (**PWR** and **USB** connectors, **JST** connector, **LED**, **LDO**, and the **DC-DC** converter 5VDC → 3.3VDC)
- **Yellow** highlights the **mikroBUS™** connector (slot **M1**), the **RPI40** and **AUX** connectors
- **Blue** highlights the elements related to the **RS485** implementation
- **Light blue** highlights the 40-pin connector for stacking with **Raspberry Pi** (mounted on the bottom-side)
- **Pink** highlights the 12-pin connectors carrying the mezzanine signals
- **Purple** highlights the **Qwiic©** connectors for connecting peripherals over the I^{2}C bus
- **Orange** highlights the **EEPROM**
- **Green** highlights the **Watchdog**

Below is the layout of the elements on the bottom-side of the **IoTbase RPi** module:



In this picture:

- Light blue highlights the 40-pin connector for stacking with Raspberry Pi

JUMPERS

All **IoTbase RPi** module jumpers are located on the bottom-side:

1. **SB1, SB2, SB3 and SB4**. Designed to select a direct or cross connection of the UART signals between **Raspberry Pi** and the module. Not used in this version of the module.
2. **SB5**. Connects the **GPIO4** signal of **Raspberry Pi** to the **RSE** circuit, used in the software-controlled direction control mode of **RS485** (see the **RS485** section). Required for software-controlled mode together with jumpers **JEN** and **JDI** in position 1-2. By default — open.
3. **SB6**. Enables the 120Ω termination resistor (**R16**) of the **RS485** bus between lines A and B. Close only if the module is installed at the physical end of the **RS485** bus (see the **RS485** section). By default — open.
4. **SB9 and SB11**. For connecting pull-up resistors to the **SDA1** and **SCL1** signals of the I^2C1 bus of the **IoTbase RPi** module. By default — open.
5. **SB10**. Connects the reset signal **RST1** (watchdog output, U8 **TPS3828-33DBV**) to the **AUX** connector (pin3), for an external hardware reset of **Raspberry Pi** via the **RUN** contacts (externally, with a separate wire). Also connects **RST1** to **GPIO23**. By default — open.
6. **SB12, SB13 and SB14**. For selecting the address of the **EEPROM** chip on the I^2C bus. Default address – 1010111x (all jumpers open). The table for selecting the address is provided below.

SB12	SB13	SB14	EEPROM address
CLOSE	CLOSE	CLOSE	1010000x
CLOSE	CLOSE	OPEN	1010001x
CLOSE	OPEN	CLOSE	1010010x
CLOSE	OPEN	OPEN	1010011x
OPEN	CLOSE	CLOSE	1010100x
OPEN	CLOSE	OPEN	1010101x
OPEN	OPEN	CLOSE	1010110x
OPEN	OPEN	OPEN	1010111x

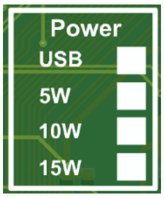
7. **SB15 and SB16**. Determine whether the module's +3V3 is connected to the **Qwiic©** connectors (**I2C-A** and **I2C-B** respectively). Allows powering external devices on the I²C bus. By default — closed, i.e. connected.
8. **SB17 and SB18**. For connecting pull-up resistors to the **SDA** and **SCL** signals of the main I²C bus of the **IoTbase RPi** module. By default — closed, i.e. connected.
9. **SB19**. **DO NOT CLOSE!**
10. **SB20**. For obtaining +5V power from **Raspberry Pi**. By default — open, i.e. not connected.
11. **SB21 and SB22**. Determine whether +3V3 and +5V, respectively, are connected to another module through the **RPI** connector. By default — both closed, i.e. connected.

CONFIGURATION TABLES

On the bottom side of the module, configuration tables provide information about the configuration of a specific module instance.
The configuration tables indicate:

The module can have **8 Kbit** or **64 Kbit EEPROM**.

The method of powering the module: via the **USB** connector or the power of the installed DC to 5VDC input voltage converter on the module. Depending on the type of converter installed, the maximum current can be 1A, 2A, or 3A, which corresponds to converter power of 5W, 10W, and 15W. It should be noted that a delivery option without any power supply method can be chosen, but in this case, 5VDC can be supplied from **Raspberry Pi** via the **SB20** or two-pin JST connector.



Whether the **IoTbase RPi** module has **RS-485** installed. CAN is not supported in the current version.

MEZZANINES

Any mezzanine from the **IoTextra** series listed on the website **http://www.makethingshappy.io** can be used as a mezzanine.
The following figures show the **IoTbase RPi** module with different mezzanines:

		
IoTbase RPi with **IoTextra Input**	**IoTbase RPi** with **IoTextra Octal3**	**IoTbase RPi** with **IoTextra Analog**

ACCESSORIES

The following accessories may be required to use the module:
- Mating (plug) two-pin connector for module power
- Mating (plug) three-pin connector for RS-485
- 12-pin FFC cable for HOST-PF connector
- 40-pin FFC cable for RPI40 connector
- **I²C cable with Qwiic©** connectors on both ends of the cable